

Design Requirements

Goal:

- Design a throttleable liquid bipropellant engine capable of lifting, hovering, and controlling a minimal VTVL hopper vehicle — a structure that exists solely to support the engine's required subsystems. No payload, no altitude target.

What does success look like?

- Simulate stable combustion for the full burn duration (60 sec target) at 100%, 66%, and 33% thrust. No instabilities at any throttle point. Structural margins positive at all operating conditions. Confirm $\frac{T}{W} > 1.0$ at hover throttle setting for the expected vehicle mass.

Requirements:

- Design if built be under \$5000 (not including fuel)
- Pressure-fed
- No cryogenics
- Design the system to be stable such that any possible instability is found and avoided within a minimum margin of 20%
- Materials must be machinable and readily accessible such as aluminum or stainless steel
- Standard AN/MS hardware and Swagelok-compatible tube fittings only. No custom fluid connectors
- All pressure-wetted components (chamber, tanks, lines) shall be designed to a burst factor of safety $\geq 4 \times$ MEOP (Maximum Expected Operating Pressure)
- Throttle range 30 $\rightarrow$ 100% of full thrust. Minimum throttle must be below hover $\frac{T}{W} = 1.0$ for the expected vehicle mass, so the engine can command descent. O/F ratio shall remain within $\pm 15\%$ of optimum across the full throttle range.
- Specific impulse $I_{sp} \geq 230$ s sea level (hard minimum). Target $\sim 250 \rightarrow 260$ s depending on propellant selection. To be computed via CEA at optimum $\frac{O}{F}$ and $P_c = 150$ psi baseline.
- Engine dry mass target ≤ 2.5 kg (chamber + nozzle + injector + cooling jacket). Target 1.0 $\rightarrow$ 1.5 kg.
- TVC interface: Engine assembly shall include a defined gimbal hard point or flange interface for future TVC integration. Geometry TBD, but the interface must not be retroactively blocked by cooling jacket or plumbing routing.

Derived Requirements:

- 400 N (90 lbf) nominal. Acceptable range: 300 $\rightarrow$ 500 N. Final value to be confirmed after vehicle mass estimation converges. Must satisfy $\frac{T}{W} \geq 1.5$ at GLOW at full throttle.

Rationale: Sized to hover a 10–20 kg vehicle with throttle margin. At 400 N on a 15 kg vehicle, $\frac{T}{W} = 2.7$ at full thrust; hover throttle is 37% ($148\text{ N} \approx 15\text{ kg } g$). This is the correct hopper operating envelope — low $\frac{T}{W}$ means throttle range actually spans from climbing to descending.

- Burn time Minimum 30 seconds. Design target 60 seconds. At $\frac{400\text{ N}}{260\text{ s}}$ I_{sp} : 30 sec = 4.7 kg propellant; 60 sec = 9.4 kg propellant.

Rationale: 30 sec is enough to demonstrate a meaningful flight profile. 60 sec gives room for an ascent, hover, translation, and descent sequence — a proper hopper demonstration. Both are feasible at this thrust level without large tanks.

- Chamber pressure (P_c) 100 $\rightarrow$ 200 psi. Baseline: 150 psi. Lower than previous revision because smaller engine diameter means thinner walls at the same stress, and lower P_c reduces pressurant requirements, simplifying the feed system.

Rationale: The performance penalty of lower P_c is small: dropping from 200 to 150 psi costs roughly 3–5 s of Isp. For a hopper where propellant mass is 5–10 kg anyway, this has negligible impact on mission duration. Simpler feed system and cheaper valves are worth more than those 3–5 seconds.

Propellant Choice

Engine Archetypes

Thermochemical Analysis

Engine Sizing

Nozzle Design

Injector Design

Feed System Design

Cooling Analysis

Structural Analysis

CAD Design

Simulation